SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05 **Total Marks:50**

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding *(2.5)*	Constraint Analysis *(2.5)*	Solution Development *(2.5)*	Application of Concepts *(1.5)*	Justification *(1)*	***Total(10)***
Weightage	*25%*	25%	25%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _K.Nagapreetham(192525242)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID	**Symptoms Observed**
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:
Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints.

Q2. Dataset: Online Learning Platform

Student ID **Courses Enrolled**

S1 Python, SQL

S2 Python, Data Science

S3 SQL, Machine Learning

S4 Python, SQL, Data Science S5

Python, Machine Learning

Question:
Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

Q3. Dataset: Library Borrowing Records

Transaction ID **Books Borrowed**

T1 Data Mining, Python Programming

T2 Data Mining, Database Systems

T3 Python Programming, Database Systems

T4 Data Mining, Python Programming, Database Systems T5 Data

Mining, Artificial Intelligence

Question:
Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Q4. Dataset: University Course Hierarchy

Student ID **Courses Taken**

S1 Engineering, Computer Science, Machine Learning

S2 Engineering, Information Technology

S3 Engineering, Computer Science

S4 Engineering, Computer Science, Machine Learning S5

Engineering, Information Technology

Question:
Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules.

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:
Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Q1. Hospital Patient Records

Given

Patient Symptoms

P1 Fever, Cough

P2 Fever, Headache

P3 Cough, Headache

P4 Fever, Cough, Headache

P5 Fever, Fatigue

- Number of transactions = **5**
- Minimum Support = **50%**
- Minimum Confidence = **70%**
- Constraint: **Every rule must contain Fever**.

Formulas

Support:

$$Support(X \to Y) = \frac{\text{Number of transactions containing } X \cup Y}{\text{Total number of transactions}} \times 100$$

Confidence:

$$Confidence(X \to Y) = \frac{Support(X \cup Y)}{Support(X)} \times 100$$

Rule 1: Fever → Cough

Fever and Cough occur together in:

- P1
- P4

Therefore,

$$Count(Fever, Cough) = 2$$

Support

$$Support(Fever \to Cough) = \frac{2}{5} \times 100$$

$$= \boxed{40\%}$$

Minimum required:

$$50\%$$

Since:

$$40\% < 50\%$$

❌ **Rule does not satisfy support.**

Confidence:

Confidence

Fever occurs in P1, P2, P4 and P5.

$$Count(Fever) = 4$$

Therefore,

$$Confidence(Fever \to Cough) = \frac{2}{4} \times 100$$

$$= \boxed{50\%}$$

Minimum confidence:

$$70\%$$

Since:

$$50\% < 70\%$$

❌ **Rule is invalid.**

Rule 2: Fever → Headache

Fever and Headache occur together in:

- P2
- P4

Therefore,

$$Count(Fever, Headache) = 2$$

Support

$$Support(Fever \to Headache) = \frac{2}{5} \times 100$$

$$= \boxed{40\%}$$

Since:

$$40\% < 50\%$$

❌ Does not satisfy minimum support.

Confidence:

Fever occurs 4 times.

$$Confidence(Fever \rightarrow Headache) = \frac{2}{4} \times 100$$

$$= \boxed{50\%}$$

Since:

$$50\% < 70\%$$

❌ Does not satisfy minimum confidence.

Final Answer Q1

Rule	Support Calculation	Support	Confidence Calculation	Confidence	Result
Fever → Cough	2/5 × 100	40%	2/4 × 100	50%	❌
Fever → Headache	2/5 × 100	40%	2/4 × 100	50%	❌

Q2. Online Learning Platform

The dataset is:

Student	Courses
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Minimum support: 40%

Constraint:

Every itemset must contain at least **two courses**.

Formula:

$$Support(X) = \frac{\text{Number of transactions containing } X}{\text{Total transactions}} \times 100$$

Total transactions:

$$N = 5$$

Minimum support count:

$$Minimum\ Support\ Count = 40\% \times 5$$

$$= 0.40 \times 5$$

$$= \boxed{2}$$

Therefore, an itemset must occur in **at least 2 students**.

Itemset 1: {Python, SQL}

Occurs in:

- S1
- S4

Count = 2

$$Support(Python, SQL) = \frac{2}{5} \times 100$$

$$= \boxed{40\%}$$

Since:

$$40\% \geq 40\%$$

✅ **Frequent itemset**

Itemset 2: {Python, Data Science}

Occurs in:

- S2
- S4

Count = 2

$$Support(Python, DataScience) = \frac{2}{5} \times 100$$

$$= \boxed{40\%}$$

Since:

$$40\% \geq 40\%$$

✅ **Frequent itemset**

Itemset 3: {SQL, Data Science}

Occurs only in S4.

Occurs only in S4.

$$Support(SQL, DataScience) = \frac{1}{5} \times 100$$

$$= \boxed{20\%}$$

$$20\% < 40\%$$

❌ Not frequent.

Itemset 4: {SQL, Machine Learning}

Occurs only in S3.

Occurs only in S3.

$$Support = \frac{1}{5} \times 100$$

$$= \boxed{20\%}$$

❌ Not frequent.

Itemset 5: {Python, Machine Learning}

Occurs only in S5.

$$Support = \frac{1}{5} \times 100$$

$$= \boxed{20\%}$$

❌ Not frequent.

Three-itemset

Consider:

$$\{Python, SQL, DataScience\}$$

Occurs only in S4.

$$Support = \frac{1}{5} \times 100$$

$$= \boxed{20\%}$$

❌ Not frequent.

Final Answer :

$$\boxed{\{Python, SQL\}, \quad \{Python, DataScience\}}$$

Both have:

$$\boxed{Support = 40\%}$$

The constraint of at least two courses eliminates single-course itemsets and reduces the number of candidates considered during mining.

Q3. Library Borrowing Records

Dataset:

Transaction	**Books**
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Constraint:

$$Maximum\ Itemset\ Size = 3$$

Formula

$$Support(X) = \frac{Count(X)}{Total\ Transactions} \times 100$$

Total transactions:

$$N = 5$$

1-Itemsets

{Data Mining}

Occurs in:

$$T1, T2, T4, T5$$

Count = 4

$$Support = \frac{4}{5} \times 100$$

$$= \boxed{80\%}$$

{Python Programming}

Occurs in:

$$T1, T3, T4$$

Actually, count = **3**.

$$Support = \frac{3}{5} \times 100$$

$$= \boxed{60\%}$$

{Database Systems}

Occurs in:

$$T2, T3, T4$$

Count = 3.

$$Support = \frac{3}{5} \times 100$$

$$= \boxed{60\%}$$

{Artificial Intelligence}

Occurs in:

$$T5$$

Count = 1.

$$Support = \frac{1}{5} \times 100$$
$$= \boxed{20\%}$$

2-Itemsets

{Data Mining, Python Programming}

Occurs in:

$$T1, T4$$

Count = 2.

$$Support = \frac{2}{5} \times 100$$
$$= \boxed{40\%}$$

{Data Mining, Database Systems}

Occurs in:

$$T2, T4$$

$$Support = \frac{2}{5} \times 100$$
$$= \boxed{40\%}$$

{Python Programming, Database Systems}

Occurs in:

$$T3, T4$$

$$Support = \frac{2}{5} \times 100$$
$$= \boxed{40\%}$$

{Data Mining, Artificial Intelligence}

Occurs in T5.

$$Support = \frac{1}{5} \times 100$$

$$= \boxed{20\%}$$

3-Itemset

{Data Mining, Python Programming, Database Systems}

Occurs only in:

$$T4$$

Therefore:

$$Support = \frac{1}{5} \times 100$$

$$= \boxed{20\%}$$

Final Table:

Itemset	**Count**	**Support**
{Data Mining}	4	80%
{Python Programming}	3	60%
{Database Systems}	3	60%
{Artificial Intelligence}	1	20%
{Data Mining, Python Programming}	2	40%
{Data Mining, Database Systems}	2	40%
{Python Programming, Database Systems}	2	40%
{Data Mining, Artificial Intelligence}	1	20%
{Data Mining, Python Programming, Database Systems}	1	20%

Important: The question specifies the maximum itemset size as 3 but **does not provide a minimum support threshold**. Therefore, we should not invent one. The table above gives the calculated supports.

Computational Complexity Reduction

Without the constraint, Apriori could generate larger combinations.

With: $|I| \leq 3$

all itemsets containing 4 or more items are immediately excluded.

Thus, candidate generation and support counting are reduced.

Q4. University Course Hierarchy

Dataset:

Student	Courses
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

The required level is: Computer Science

Step 1: Support of Computer Science

Computer Science occurs in:

$$S1, S3, S4$$

Count = 3.

Formula

$$Support(CS) = \frac{Count(CS)}{Total\ Students} \times 100$$

Substitution

$$= \frac{3}{5} \times 100$$

$$= \boxed{60\%}$$

Step 2: Support of Machine Learning

Machine Learning occurs in:

$$S1, S4$$

Count = 2.

$$Support(ML) = \frac{2}{5} \times 100$$
$$= \boxed{40\%}$$

Step 3: Support of Computer Science + Machine Learning

Both occur together in:

$$S1, S4$$

Count = 2.

$$Support(CS, ML) = \frac{2}{5} \times 100$$
$$= \boxed{40\%}$$

Rule 1: Computer Science → Machine Learning

Formula

$$Confidence(CS \to ML) = \frac{Support(CS, ML)}{Support(CS)} \times 100$$

Substitution

$$= \frac{40}{60} \times 100$$
$$= \boxed{66.67\%}$$

Therefore:

$$\boxed{CS \to ML}$$

has:

- Support = **40%**
- Confidence = **66.67%**

Rule 2: Machine Learning → Computer Science

Formula

$$Confidence(ML \rightarrow CS) = \frac{Support(ML, CS)}{Support(ML)} \times 100$$

Substitution

$$= \frac{40}{40} \times 100$$

$$= \boxed{100\%}$$

Therefore:

$$\boxed{ML \rightarrow CS}$$

has:

- Support = **40%**
- Confidence = **100%**

Final Q4 Answer

Association Rule	Support	Confidence
Computer Science → Machine Learning	40%	66.67%
Machine Learning → Computer Science	40%	100%

Interpretation: Every student who takes Machine Learning in this dataset also takes Computer Science.

The hierarchy constraint limits rule generation to the specified **Computer Science level**, avoiding unnecessary rules from other course categories.

Q5. Insurance Customer Records

Dataset:

Customer	Age	Income	Policy
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance

Customer	Age	Income	Policy
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Constraint:

$$Age\ Group = Young$$

Therefore, first filter the dataset.

Young Customers

C1, C3 and C5.

$$Total\ Young = 3$$

Pattern 1: Young + High Income → Health Insurance

Young customers with High income:

C1.

Count: 1

Support:

$$Support(Young, High, Health) = \frac{1}{3} \times 100$$

$$= \boxed{33.33\%}$$

Confidence

For the filtered Young group:

$$nfidence(Young + High \rightarrow Health) = \frac{Count(Young, High, Health)}{Count(Young, High)} \times 100$$

$$= \frac{1}{1} \times 100$$

$$= \boxed{100\%}$$

Pattern 2: Young + Low Income → Travel Insurance

Young + Low + Travel occurs for C3.

Count: 1

Support

$$Support = \frac{1}{3} \times 100$$

$$= \boxed{33.33\%}$$

Confidence

$$Confidence(Young + Low \to Travel) = \frac{1}{1} \times 100$$

$$= \boxed{100\%}$$

Pattern 3: Young + Medium Income → Vehicle Insurance

Young + Medium + Vehicle occurs for C5.

Count: 1

Support

$$Support = \frac{1}{3} \times 100$$

$$= \boxed{33.33\%}$$

Confidence

$$Confidence(Young + Medium \to Vehicle) = \frac{1}{1} \times 100$$

$$= \boxed{100\%}$$

Final Table:

Pattern	Substitution	Support	Confidence
Young + High → Health	1/3 × 100	33.33%	1/1 × 100 = 100%
Young + Low → Travel	1/3 × 100	33.33%	1/1 × 100 = 100%
Young + Medium → Vehicle	1/3 × 100	33.33%	1/1 × 100 = 100%

Marketing Interpretation:

The patterns can help divide young customers according to income:

$$\boxed{High\ Income \to Health\ Insurance}$$

$$\boxed{Low\ Income \to Travel\ Insurance}$$

$$\boxed{Medium\ Income \to Vehicle\ Insurance}$$

This allows an insurance company to design different offers for different income groups.

Note: Because the dataset contains only three young customers, these patterns should be treated as observations from the given dataset, not as strong conclusions about the entire population.